Step 2b - H1 - Pharma continuous solvent removal

Tech baseline

Technologies used in the market

Batch fluid-bed dryer (hot-air convection)

How it works: Heated process air fluidizes wet granules in a bowl; solvent/water evaporates from the particle surface into the airstream.

How well it performs: Typical 30-90 min per batch, inlet air 40-80 C; standard workhorse for aqueous granulation; for solvents needs closed N2 loop + ATEX rating, increasing capex and operating cost.

Limits: Heat reaches the core only by conduction from the surface, so drying rate falls off steeply as the surface dries; batch-only; large air volumes make solvent recovery expensive.

Segmented semi-continuous fluid-bed dryer (GEA ConsiGma GD class)

How it works: Wet granules from a twin-screw granulator are filled sequentially into drying cells; each cell convectively dries a small batch, giving quasi-continuous flow at line level.

How well it performs: 5-120 kg/h throughput; per-cell drying on the order of minutes to ~20 min; commercial reference for continuous OSD drying.

Limits: Still convective - solvent versions need full inert loop; moisture endpoint relies on soft sensors rather than direct in-line measurement in every cell; granule breakage/attrition in transfer documented in literature.

Truly continuous fluid-bed transport dryer (L.B. Bohle QbCon dryer)

How it works: Granules are conveyed through a plug-flow fluidized dryer with narrow residence-time distribution for traceability and targeted rejection.

How well it performs: QbCon 1 R&D scale 0.5-2.5 kg/h; QbCon line 5-25 kg/h; uniform drying demonstrated over multi-hour runs; only 'true continuous' dryer marketed for pharma R&D.

Limits: Convective heat transfer limits drying speed; marketed primarily for aqueous granulation; still at R&D/small-production scale.

Vacuum tray dryer

How it works: Static trays in a heated vacuum chamber; heat by conduction through the shelf, vapour removed by vacuum.

How well it performs: Low-temperature, gentle, closed solvent recovery; typical cycles 4-24 h.

Limits: Very slow - poor heat transfer into a static bed; purely batch; labour-intensive loading/unloading; not a candidate for continuous lines.

Single-pot processor with microwave-vacuum drying (GEA UltimaPro, L.B. Bohle VMA)

How it works: High-shear mixer-granulator whose bowl is evacuated; microwave energy volumetrically heats the wet granule bed while vacuum lowers the boiling point; vapour condensed for solvent recovery.

How well it performs: Production-proven in GMP (e.g., UltimaPro 600 batch); microwave option cuts drying time ~2-4x vs vacuum alone; handles organic-solvent granulation with

closed solvent recovery.

Limits: Batch operation only - drying happens in the granulation vessel between batches; this is the closest existing proof that MW-vacuum drying works in pharma, but it does not deliver continuous flow.

Continuous drum coater with hot-air drying (Driam Driaconti-T, O'Hara class)

How it works: Tablets move continuously through a rotating perforated drum while coating solution is sprayed and evaporated by hot process air.

How well it performs: Continuous coating at production scale; integrates in Glatt MODCOS lines; solvent versions exist with ATEX design and N2 circuits.

Limits: Drying remains air-limited, forcing lower spray rates for solvent films; residence-time spread is wide; not a standalone solvent-removal module for granules.

Why our solution can be better

Microwave energy couples directly into the polar solvent throughout the wet granule bed, so drying no longer depends on slow surface-first convective heat transfer; under vacuum the solvent boils at low temperature, protecting heat-sensitive APIs. Published lab work shows microwave-vacuum drying of pharma granules roughly 3-10x faster than convective or vacuum-only drying, and MW-vacuum is already GMP-proven in batch single-pot processors - so the credible wedge is a genuinely continuous solvent-removal module, an engineering gap rather than a science gap. The advantage is plausible but not yet demonstrated at continuous scale: field uniformity across a moving bed (penetration depth is cm-scale at 2.45 GHz), arcing/explosion control in a flammable solvent atmosphere, and a credible in-line moisture PAT story are the make-or-break items.

Key assumptions

• Microwave-vacuum drying removes solvent from pharma granules ~3-10x faster than convective/vacuum drying at comparable product temperature (measured, lab-scale studies on HCT granules: ~9-30 min vs hours)

• Polar solvents used in pharma granulation/coating (ethanol, IPA, acetone, water) absorb strongly at 2.45 GHz, enabling selective volumetric heating of the wet phase (measured, dielectric literature)

• Required throughput class is ~5-120 kg/h to bolt onto ConsiGma/QbCon-class continuous lines (measured, competitor specs)

• Only ~10-20% of OSD granulation/coating processes still use organic solvents - the rest moved to aqueous since the 1990s, capping niche size (assumed)

• An ATEX-compliant, N2-inerted continuous MW applicator with condenser-based solvent recovery is feasible (assumed - the central engineering risk)

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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis